```
auto_failback on
node rproxy1 rproxy2
```

The explanation of these options is given below:

- *use_logd* specifies the use of the logging daemon to log all the messages. This option has deprecated the debugfile/logfile/logfacility log options. Using this is recommended.
- *keepalive* specifies the number of seconds between two heartbeats.
- *deadtime* specifies the number of seconds after which a host is considered dead if not responding.
- *warntime* specifies the number of seconds after which the late Heartbeat warning will be issued.
- *initdead* is the number of seconds to wait for the other host after starting Heartbeat, before it is considered dead.
- *udpport* is the port number used for bcast/ucast communication. The default value for this is 694, but I've used 695 because I had a pair of HA LM-1500 load balancing appliances from Kemp Technologies on the same subnet for some Citrix servers. This appliance is based on Linux and to me it looked like it was using Heartbeat for HA on the default UDP port 694.
- *bcast/ucast* is the interface on which to broadcast/unicast. If you are planning to make this only a two-node cluster, then there is no need for sending broadcasts; instead, use unicast. Now you will see that one of the IP addresses to which unicast is being sent is the local machine itself. I have done this to make sure that the *ha.cf* file is identical on both the cluster nodes. The unicast directives sent to the local machines are effectively ignored.

Note: If you have changed the default UDP port (like I have done above) then make sure that the 'ucast' or 'bcast' line is after the 'udpport' line; otherwise, the default port 694 will be used. Remember that the order of the DIRECTIVES in *ha.cf* is important and matters.

- If *auto_failback* is set to 'on', then resources are automatically failed back to its primary node.
- The *node* directive specifies the nodes in the HA set-up. The name specified here must match with the *uname -n* of the cluster node.

The *haresources* file

Now we need to tell Heartbeat about the resources the cluster will be managing. There are two ways of doing this. One, by using the *haresources* file, and two, by enabling the Cluster Resource Manager (CRM) and using *cib.xml*. In Heartbeat 2.x versions, if CRM is enabled, *haresources* is not used.

Setting up clustering using CRM is unnecessarily complicated for our simple set-up, although the Linux-HA project provides the command line tools and GUI tools to manage the set-up. In our set-up, we will use the Heartbeat R1 style clustering (named so because of its compatibility with the older release 1.x of Heartbeat). Once you can get this working, you can move towards setting up the Heartbeat R2 style clustering.

If you are using the *haresources* file for your set-up, you need to make sure this file is identical on both the machines. The general syntax of this file is to list the preferred-node followed by the list of resources that will be running on this preferred-node. All the resources specified on a single line are called a resource-group. In order to continue on the next line, a '\' can be used. The first resource in each resource group (in case you are specifying multiple resource groups) needs to be unique because it is used as the resource-group name. The preferred-node is the one where the listed resources will run by default when both (or all) nodes of a cluster are available and the 'autofailback' option is set to 'on' in the *ha.cf* file.

Shown below is my haresources file:

```
# cat /etc/ha.d/haresources
rproxy1 10.8.0.1 nginx
```

While taking-over or acquiring the resources, Heartbeat uses left-to-right ordering and while releasing, a right-to-left order. All resources under cluster control must have a resource control script that is typically located in either */etc/init.d* or */etc/ha.d/resource.d* directories. Any script that can take at least two parameters, start and stop—to start and stop the resource, respectively—can be used as a resource control script. The IP address used in our *haresources* file is the service IP address, where our nginx reverse proxy server will be serving the requests. In DNS, this IP is to be mapped with *www.unixclinic.net*, which is what the customers will be accessing. You can see that we have not used any resource control script to acquire the IP address. The reason for this is that a service IP address is typically the requirement for all kinds of HA clusters that we set up and hence Heartbeat, by default, uses a resource control script called *IPaddr* for acquiring the IP address. So instead, the *haresources* file could have been written as:

```
rproxy1 IPaddr::10.8.0.1 nginx
```

...or if we want to specify netmask, interface or broadcast values for the service IP, then we can use the following syntax:

```
rproxy1 IPaddr::10.8.0.1/255.255.255.0/eth1/255.255.255.255
```

In our *haresources* file, we have only specified the service IP address for the cluster, but have not specified which interface in the machine will acquire this IP address; neither have we specified the netmask and broadcast values. In such cases, these values are set automatically by Heartbeat, by looking at the routing table. Heartbeat attempts to find the lowest cost route to the service IP address, and if multiple interfaces are found to provide the lowest cost route, then the first such route is considered. This basically means that the default route of the system is the least preferred. For setting up the broadcast address, the largest available address is used.

For details on this, see the side-box: 'IPaddr versus IPaddr2